



Semester: Spring 2025
Course: Knowledge-Based Systems
Lecturer: Austin Odera
Time/Days: Monday/Wednesday 7:00 AM - 8:40 AM
Venue: LAB 8, FBSC
Credit: 3 Units
Tel/Email: aoder@usiu.ac.ke/+254710249790
School of Science and Technology

Course Description

This course introduces the concepts and methodologies of Knowledge-Based Systems (KBS). Students will learn how to design, develop, and implement intelligent systems that utilize knowledge representation and reasoning techniques. The course emphasizes practical applications in expert systems, decision-making, and problem-solving.

Course Learning Outcomes

By the end of the course, students will be able to:

1. Define and explain the core concepts of Knowledge-Based Systems and artificial intelligence.
2. Design knowledge representation structures such as rules, semantic networks, and ontologies.
3. Implement reasoning mechanisms for decision-making and problem-solving.
4. Evaluate and apply KBS in real-world scenarios.

Program Learning Outcomes

Upon completion of the program, students will be able to:

1. Demonstrate an understanding of intelligent system principles and applications.
2. Solve complex problems using knowledge-based techniques.
3. Develop and evaluate the performance of intelligent systems in various domains.

Teaching Methodology and Materials

This course employs a mix of instructional strategies to promote active learning:

****Lectures: **** Discussing theoretical concepts and applications in KBS.

****Tutorials: **** Addressing problem-solving and design techniques for knowledge systems.

****Laboratory Exercises: **** Hands-on practice in developing and testing KBS solutions.

Course Content Overview

Weeks 1-2: Introduction to Knowledge-Based Systems

- Overview of Knowledge-Based Systems.

- Characteristics and Applications of KBS.

- Historical Evolution and Current Trends.

****Assignments:**** Write a report on the historical development of KBS.

****Recommended Readings:**** Chapters 1-2 of "Knowledge-Based Systems" by Akerkar, Rajendra.

Weeks 3-4: Knowledge Representation

- Types of Knowledge: Declarative and Procedural.

- Techniques: Rules, Frames, Semantic Networks, Ontologies.

- Logic-Based Representation.

****Assignments:**** Design a rule-based system for a simple decision-making scenario.

****Recommended Readings:**** Chapter 3.

Weeks 5-6: Reasoning and Inference

- Introduction to Inference Engines.

- Forward and Backward Chaining.

- Probabilistic Reasoning.

****Assignments:**** Develop a reasoning mechanism for a hypothetical problem.

****Recommended Readings:**** Chapters 4-5.

Week 7: Mid-Semester Exam

Weeks 8-9: Expert Systems Development

- Architecture of Expert Systems.

- Knowledge Acquisition Techniques.

- Case Studies of Expert Systems.

****Assignments:**** Create an outline for an expert system in a chosen domain.

****Recommended Readings:**** Chapters 6-7.

Weeks 10-11: Emerging Trends in KBS

- Integration with Machine Learning.

- Role of Big Data and Cloud Computing in KBS.

- Applications in Healthcare, Agriculture, and Finance.

****Assignments:**** Research and present on an emerging trend in KBS.

****Recommended Readings:**** Chapters 8-9.

Weeks 12-13: Evaluation and Validation of KBS

- Metrics for Performance Evaluation.

- Techniques for Testing and Validation.

- Ensuring System Scalability and Reliability.

****Assignments:**** Evaluate a KBS case study provided in class.

****Recommended Readings:**** Chapter 10.

Week 14: End-Semester Exam

Course Text

****Knowledge-Based Systems**** by Rajendra Akerkar

****Publisher:**** Jones & Bartlett Learning

****Year:**** 2010 (or latest available edition)

Class Policy

- Assignments submitted late will not be accepted.
- All assignments must be submitted via the LMS platform.
- Academic dishonesty will lead to disciplinary action.
- Students must attend at least 80% of classes to pass.

Course Evaluation

- Attendance & Participation: 5%
 - Assignments: 10%
 - Group Project: 15%
 - Lab Exercises: 10%
 - Quizzes & Tests: 15%
 - Mid-Semester Exam: 20%
 - Final Exam: 25%
- Total: 100%